13-permutation-group: 置换群 (Permutation Group)

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$$M = \{1, 2, \ldots, n\} \qquad S_n \triangleq \mathrm{Sym}(M)$$

Definition (Symmetric Group (对称群; $\mathrm{Sym}(M)$))

Let $M \neq \emptyset$ be a set.

All the permutations (排列; 置换)/bijections (双射) of M,

together with the composition operation, is a group,

called the symmetric group of M.

$$S_3$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \qquad \tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\tau\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$\sigma\tau \neq \tau\sigma$$

Cyclic Notation (轮换表示法) & Transposition (对换)

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 3 & 6 & 1 & 5 & 2 \end{pmatrix}$$

$$\begin{aligned} \sigma &= (1\ 4)(2\ 3\ 6)(5) \\ &= (1\ 4)(2\ 3\ 6) \\ &= (2\ 3\ 6)(1\ 4) \\ &= (2\ 3\ 6)(4\ 1) \\ &= (3\ 6\ 2)(4\ 1) \\ &= (3\ 6)(6\ 2)(4\ 1) \end{aligned}$$

Theorem

Every permutation can be expressed as a product of disjoint cycles.

Theorem

Disjoint cycles commute with each other.

Theorem

Every permutation can be expressed as a product of transpositions.

$$(i_1\ i_2\ \ldots\ i_r) = (i_1\ i_2)(i_2\ i_3)\ldots(i_{r-2}\ i_{r-1})(i_{r-1}\ i_r)$$

By induction on the length r.

$$S_3$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$(1) \qquad (1\ 3\ 2) \qquad (1\ 2\ 3)$$

$$(1\ 2) \qquad (1\ 3) \qquad (2\ 3)$$

$$
\begin{aligned}
\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 3 & 6 & 2 & 5 & 4 & 1 \end{pmatrix} &= (1\ 7)(2\ 3)(3\ 6)(6\ 4) \\
&= (1\ 7)(3\ 6)(2\ 5)(6\ 4)(4\ 5)(2\ 5)
\end{aligned}
$$

Theorem (Parity (奇偶性) of Permutations)

将一个置换表示成若干对换的乘积，所用对换个数的奇偶性是唯一的。

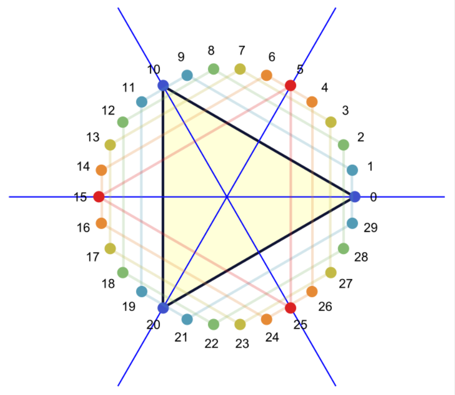
Definition (Even/Odd Permutations (偶置换/奇置换))

可表示为偶数个对换的乘积的置换称为偶置换; 否则, 称为奇置换。

Definition (Alternating Group (交错群; A_n))

由 S_n 的全体偶置换构成的子群称为 n 次交错群。

$$A_3 = \{(1), (1\ 2\ 3), (1\ 3\ 2)\}$$



$$A_3 = \{(1), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$(1\ 2)A_3 = \{(1\ 2), (2\ 3), (1\ 3)\}$$

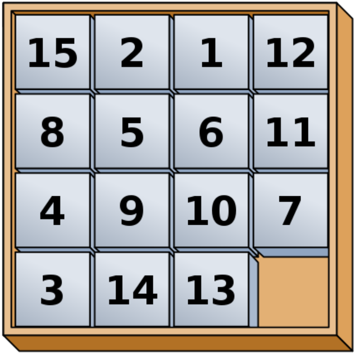
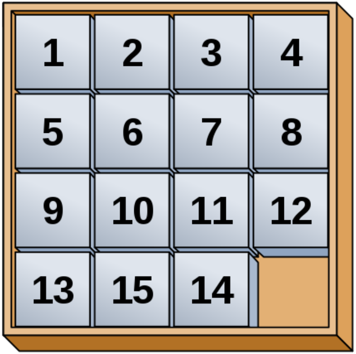
Definition (Permutation Group (置换群))

Let $M \neq \emptyset$ be a set.
A permutation group of M is a subgroup of $\mathrm{Sym}(M)$.

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_3$$


15 2 1 12
8 5 6 11
4 9 10 7
3 14 13

1 2 3 4
5 6 7 8
9 10 11 12
13 15 14

A Modern Treatment of the 15 Puzzle

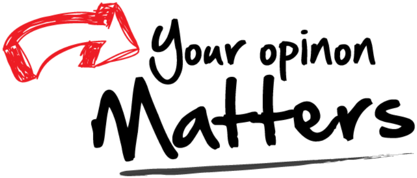
Aaron F. Archer

1. INTRODUCTION. In the 1870's the impish puzzlemaker Sam Loyd caused quite a stir in the United States, Britain, and Europe with his now-famous 15-puzzle. In its original form, the puzzle consists of fifteen square blocks numbered 1 through 15 but otherwise identical and a square tray large enough to accommodate 16 blocks. The 15 blocks are placed in the tray as shown in Figure 1, with the lower right corner left empty. A legal move consists of sliding a block adjacent to the empty space into the empty space. Thus, from the starting placement, block 12 or 15 may be slid into the empty space. The object of the puzzle is to use a sequence of legal moves to switch the positions of blocks 14 and 15 while returning all other blocks to their original positions.

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

Figure 1. The starting position for the 15-puzzle. The shaded square is left empty.


Thank
You!



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